

Prolog Program for Medical Diagnosis

1. Aim

To write and execute a Prolog program that diagnoses a probable disease based on a given set of symptoms, using facts, rules, and queries.

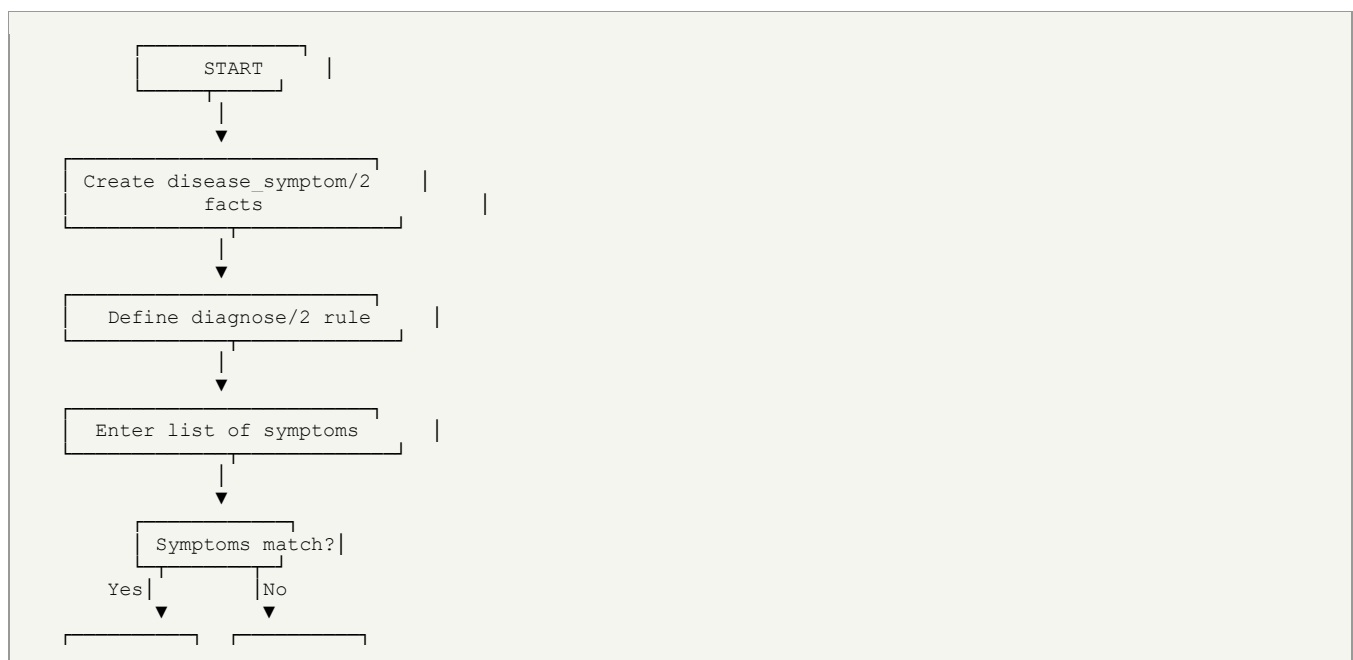
2. Objective

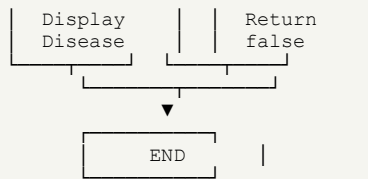
- To understand how expert-system-style diagnosis can be modelled in Prolog.
- To represent diseases as sets of associated symptoms using facts.
- To determine a matching disease by checking that all its symptoms are present in the input.
- To retrieve a diagnosis using a query with a list of observed symptoms.

3. Algorithm

1. Start the program.
2. Create a Prolog database using the predicate `disease_symptom/2`, mapping each disease to its list of symptoms.
3. Define a helper predicate `subset/2` to check that every symptom of a disease is present in the observed list.
4. Define a rule `diagnose/2` that finds a disease whose full symptom list is a subset of the observed symptoms.
5. Enter the list of observed symptoms as a query.
6. Search the database for a disease whose symptoms are all present.
7. If found, display the diagnosed disease; otherwise Prolog returns false.
8. Stop the program.

4. Flowchart





5. Code with Output

Code:

```

% -----
% Medical Diagnosis Database
% disease_symptom(Disease, [Symptoms])
% -----
disease_symptom(flu,      [fever, cough, headache]).
disease_symptom(measles,  [fever, rash]).
disease_symptom(migraine, [headache, nausea]).
disease_symptom(cold,     [cough, sneezing]).

% -----
% Subset check
% -----
subset([], _).
subset([H|T], L) :-
    member(H, L),
    subset(T, L).

% -----
% Diagnose rule
% -----
diagnose(Symptoms, Disease) :-
    disease_symptom(Disease, DiseaseSymptoms),
    subset(DiseaseSymptoms, Symptoms).

```

Output:

```

?- diagnose([fever, cough, headache], D).
D = flu.

?- diagnose([fever, rash], D).
D = measles.

?- diagnose([itchy, eyes], D).
false.

```

6. Result

The Prolog program was successfully implemented to diagnose a probable disease from a list of observed symptoms by checking which disease's full symptom set is contained in the input.

7. Conclusion

Thus, the medical diagnosis program was successfully implemented using Prolog. The experiment demonstrates a simple rule-based expert system where subset checking over symptom lists drives the diagnostic decision.